RANDOM MEASURE

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Let S be an open subset of $\mathbb{R}^d$, denote $\mathcal{B}(S)$ the Borel σ-field on it (one can generalize to S being a very general topological space).

A "random measure" on $(S, \mathcal{B}(S))$ is locally finite kernel from $\underbrace{(\Omega, \mathcal{A})}_{\text{prob. space}}$ to $(S, \mathcal{B}(S))$.

Say ξ to be the measure. Then

$\omega \longmapsto \xi(\omega, B)$ is a r.v. for fixed $B \in \mathcal{B}(S)$

$B \longmapsto \xi(\omega, B)$ is a locally finite measure on $\mathcal{B}(S)$

In other words $\xi : \Omega \times \mathcal{B}(S) \longmapsto [0, +\infty]$

(locally finite means $\xi(\omega, C) \underset{\text{a.s.}}{<} +\infty$ C is rel-compact)

Typical notation:

$$\xi B := \xi(\cdot, B) \qquad \xi f = \int f \, d\xi = \int f(s) \, d\xi(\cdot, ds)$$

NB: the integral ξf is measurable ... hence a r.v., for every $f : S \mapsto \mathbb{R}^+$ measurable.

NB $\quad E\xi$ given by $(E\xi)(B) = E(\xi B)$ is a measure!!

$$\boxed{= \int_\Omega \int 1_B(s) \, \xi(\omega, ds) \, dP}$$

$$\boxed{= \int_\Omega \xi(\omega, B) \, dP}$$

Recall theory of kernels

$E\xi$ is a measure!!

LEMMA. Let ξ and η be random measures on S. Then $\xi \overset{d}{=} \eta$ iif $E e^{-\xi f} = E e^{-\eta f}$ for all $f \in C_c^+(S)$ and also iif

$$(\xi B_1, \dots, \xi B_m) \overset{d}{=} (\eta B_1, \dots, \eta B_m)$$

$B_1 \dots B_m$ rel-compact. $m \in \mathbb{N}$

NB: It is equivalent to think of ξ as a random element in the space $\mathcal{M}(S)$ of locally finite measures on S, equipped with the σ-field induced by all the evaluation maps $\pi_B : \mu \longmapsto \mu B$, i.e., $\sigma(\pi_B, B \in \mathcal{B}(S))$

$\mathcal{M}(S) \longmapsto [0, +\infty]$

$\hookrightarrow$ for stoch. processes this was $\sigma(\pi_t, t \geq 0)$

Hence, two random measures have the same law if their laws on $\sigma(\pi_B, B \in \mathcal{B}(S))$ are the same.

By a point process on S we mean an integer valued random measure ξ. In this case ξB is clearly a $\mathbb{Z}_+$-valued r.v., for every B compact.

A random measure ξ on S is said to have <u>independent increments</u> if $\xi B_1 \dots \xi B_m$ are indep for every disjoints $B_1 \dots B_m$ rel-compacts, $m \in \mathbb{N}$.

By a <u>POISSON PROCESS</u> on S with intensity measure $\mu \in \mathcal{M}(S)$ we mean a point process ξ with independent increments such that ξB has Poisson distribution with mean μB for every B rel. compact.

$$P(\xi B = k) = \frac{(\mu B)^k}{k!} e^{-\mu B} \qquad (B \text{ rel. compact})$$

Exercise: check that these conditions specify the law of the measure. Use previous lemma.

LEMMA. Let ξ be a Poisson process on S with intensity μ. Then, for any measurable $f : S \mapsto \mathbb{R}_+$

$$E e^{-\xi f} = e^{-\mu(1 - e^{-f})} \qquad *$$

If $f : S \mapsto \mathbb{R}$ is measurable with $\mu(|f| \wedge 1) < +\infty$ then $*$ is true with $f \mapsto if$

Proof Recall that, if X is a Poisson r.v. with mean c

$$E e^{aX} = e^{-c} \sum_n \frac{(c e^a)^n}{n!} = e^{-c(1-a)}$$

Take

$$f = \sum_{k \leq m} a_k 1_{B_k} \qquad a_1 \dots a_m \in \mathbb{C} \quad B_1 \dots B_m \text{ disj. rel compact}$$

$$E e^{-\xi f} = E e^{-\sum_k a_k \xi B_k} = \prod_k E e^{-a_k \xi B_k}$$

integral of a simple function $\qquad$ INDEP. INCREM.

$$= \prod_k e^{-\mu B_k (1 - e^{-a_k})}$$

$\hookrightarrow \xi B_k$ is Poisson

$$= \exp\left\{-\sum_k \mu B_k (1 - e^{-a_k})\right\}$$

$$= e^{-\mu(1 - e^{-f})}$$

For general $f \geq 0$, choose $f_n \uparrow f$ and use monotone convergence to say

$$\xi f_n \to \xi f$$

$$\mu(1 - e^{-f_n}) \to \mu(1 - e^{-f})$$

Then for $E e^{-\xi f_n}$ note that

$$|e^{-\xi f_n}| \leq 1 \quad \text{and use DOM CONV}$$

Assume now $\mu(|f| \wedge 1) < +\infty$. Use $*$ for $c|f|$ and let $c \downarrow 0$, then

$$\underbrace{E e^{-\xi c|f|}}_{c \downarrow 0 \; P(\xi |f| < +\infty) \text{ by prop. of LT}} = \underbrace{e^{-\mu(1 - e^{-c|f|})}}_{c \downarrow 0 \; 1}$$

$|1 - e^{-cf}| \leq c|f| \wedge 2$ since $c \downarrow 0$ take $c_n \in$ comp. and use DOM CONV

For simple functions $f_n \to f$ $|f_n| \leq f$, using that

$$|1 - e^{-if_n}| \leq 2 \wedge |f|$$

Hence

$$\xi f_n \to \xi f \text{ by DOM CONV (since } \xi |f| < +\infty, \ f_n \leq |f|)$$

and

$$\mu(1 - e^{-if_n}) \to \mu(1 - e^{-if})$$

Hence

$$E e^{-i\xi f_n} \to E e^{-i\xi f}$$

and the result follows.

CONSTRUCTION OF POISSON P. PROCESSES

Take a prob. meas on S, say μ. Let $\gamma_1 \gamma_2 \dots$ be iid r.v.'s in S with distribution μ. A point process with the same distribution of

$$\xi = \sum_{k \leq m} \delta_{\gamma_k}$$

is called a sample process based on μ and m.

Take now another random variable $K \perp (\gamma_n)_n$ with values in $\mathbb{Z}^+$. A point process distributed as

$$\xi = \sum_{k \leq K} \delta_{\gamma_k}$$

is called a mixed sample process based on μ and K.

PROPOSITION

Let ξ be a point process on $(S, \mathcal{B}(S), \mu)$. Then ξ is Poisson with intensity μ iff, $\forall B \in \mathcal{B}(S)$ s.t. $\mu B \in (0, \infty)$ the restriction $1_B \cdot \xi$ is a mixed sample process based on $\mu[\cdot | B]$ and the Poisson distribution with mean $\mu[\cdot | B]$

<u>DIMO</u>

Fix $B \in S$, $0 < \mu B < +\infty$, define

$$\eta = \sum_{k \leq K} \delta_{\gamma_k} \qquad \gamma_1 \gamma_2 \dots \text{ iid law } \mu(\cdot | B)$$

and K is Poisson, indep of (γ_n), with mean μB.

Then

$$E s^K = e^{-\mu B(1-s)}$$

and so

$$E e^{-\eta f} = E e^{-\int f \, d\eta} = E e^{-\sum_{k \leq K} f(\gamma_k)}$$

$$= E\left(E e^{-\sum_{k \leq K} f(\gamma_k)} \Big| K\right)$$

$$= E\left(\prod_{k \leq K} E e^{-f(\gamma_k)} \Big| K\right)$$

$$= E\left(E e^{-f(\gamma_1)}\right)^K$$

$$= e^{-\mu B(1 - E e^{-f(\gamma_1)})}$$

$$= e^{-\mu B \int (1 - e^{-f(s)}) \frac{\mu(ds \cap B)}{\mu(B)}}$$

$$= e^{-\int (1 - e^{-f(s)}) \mu(ds \cap B)}$$

$$= e^{-\int_B (1 - e^{-f(s)}) \mu(ds)}$$

We proved before that, for a Poisson process on S with int. μ

$$E e^{-\xi f} = e^{-\mu(1 - e^{-f})}$$

hence

$$E e^{-\xi 1_B f} = e^{-\mu(1 - e^{-f 1_B})}$$

$$= e^{-\int (1 - e^{-f(s) 1_B(s)}) \mu(ds)}$$

$$= e^{-\int_B (1 - e^{-f(s)}) \mu(ds)}$$

Since the two Laplace functionals coincide they have the same law.

Theorem. For any σ-finite measure space $(S, \mathcal{S}, \mu)$ there exists a Poisson process ξ on S with $E\xi = \mu$.

PROOF. Decompose S into disjoint $B_1 B_2 \dots \in \mathcal{S}$ with measures $\mu B_n \in (0, +\infty)$. Construct independent mixed sample processes $\xi_1 \xi_2 \dots$ on S such that ξ_n is based on $\mu(\cdot | B_n)$ and the Poisson law with $E\xi_n = 1_{B_n} \cdot \mu$

Then $\xi = \sum_n \xi_n$

is Poisson with intensity $E\xi = \sum_n 1_{B_n} \mu = \mu$

Indeed

$$E e^{-\xi f} = E e^{-\sum_n \xi_n} = \prod_n E e^{-\xi_n}$$

$$= \prod_n e^{-\int_{B_n} (1 - e^{-f(s)}) \mu(ds)}$$

$$= e^{-\int_S (1 - e^{-f(s)}) \mu(ds)}$$